Answers learning task 8

2026 – 2027

Exercise 5.13: Determine the value of c such that the function $f(x, y) = cxy$ for $0 < x < 3$ and $0 < y < 3$ satisfies the properties of a joint probability density function. Determine the following:

(a) $P(X < 2, Y < 3)$
(b) $P(X < 2.5)$
(c) $P(1 < Y < 2.5)$
(d) $P(X > 1.9, 1 < Y < 2.5)$
(e) $E(X)$
(f) $P(X < 0, Y < 4)$
(g) Marginal probability distribution of the random variable X
(h) Conditional probability distribution of Y given that $X = 1.5$
(i) $E(Y|X = 1.5)$
(j) $P(Y < 2|X = 1.5)$
(k) Conditional probability distribution of X given that $Y = 2$

Answer

$c = \frac{4}{81}$

(a) 0.4444
(b) 0.6944
(c) 0.5833
(d) 0.3494
(e) 2
(f) 0
(g) $f_X(x) = \frac{2}{9}x$, for $0 < x < 3$
(h) $f_{Y|1.5}(y) = \frac{2}{9}y$, for $0 < y < 3$
(i) 2
(j) $4/9$
(k) $f_{X|2}(x) = \frac{2}{9}x$, for $0 < x < 3$

Exercise 5.15: Determine the value of c that makes the function $f(x, y) = cxy$ a joint probability density function over the range $0 < x < 3$ and $0 < y < x$. Determine the following:

(a) $P(X < 1, Y < 2)$
(b) $P(1 < X < 2)$
(c) $P(Y > 1)$
(d) $P(X < 2, Y < 2)$
(e) $E(X)$
(f) $E(Y)$
(g) Marginal probability distribution of X
(h) Conditional probability distribution of Y given $X = 1$
(i) $E(Y|X = 1)$
(j) $P(Y > 2|X = 1)$
(k) Conditional probability distribution of X given that $Y = 2$

Answer

$c = \frac{8}{81}$

(a) $\frac{1}{81}$
(b) $\frac{5}{27}$
(c) 0.7901
(d) $\frac{16}{81}$
(e) 2.4
(f) 1.6
(g) $f(x) = \frac{4}{81}x^3$, for $0 < x < 3$
(h) $f_{Y|1}(y) = 2y$, for $0 < y < 1$
(i) 0.6667
(j) 0
(k) $f_{X|2}(x) = \frac{2}{5}x$, for $2 < x < 3$

Exercise 5.27: The weights of adobe bricks used for construction are normally distributed with a mean of 1.5 kg and a standard deviation of 0.3 kg. Assume that the weights of the bricks are independent and that a random sample of 20 bricks is selected.

(a) What is the probability that all the bricks in the sample exceed 1.2 kg?

(b) What is the probability that the heaviest brick in the sample exceeds 1.8 kg?

Answer

(a) 0.0316

(b) 0.9684

Exercise 5.35: Determine the value for c and the covariance and correlation for the joint probability density function $f_{XY}(x, y) = cxy$ over the range $0 < x < 2$ and $0 < y < x$.

Answer

$c = \frac{1}{2}, \quad \text{cov}(X, Y) = 0.0711, \quad \rho(X, Y) = 0.4924$

Exercise 5.47: Suppose that X and Y have a bivariate normal distribution with $\sigma_X = 0.04$, $\sigma_Y = 0.08$, $\mu_X = 3.00$, $\mu_Y = 8.00$, and $\rho = 0$. Determine the following:

(a) $P(2.95 < X < 3.05)$
(b) $P(7.60 < Y < 7.80)$
(c) $P(2.95 < X < 3.05, 7.60 < Y < 7.80)$

Answer

(a) 0.7887
(b) 0.0062
(c) 0.0049

Exercise 5.49: In the manufacture of electroluminescent lamps, several different layers of ink are deposited onto a plastic substrate. The thickness of these layers is critical if specifications regarding the final color and intensity of light are to be met. Let X and Y denote the thickness of two different layers of ink. It is known that X is normally distributed with a mean of 0.1 millimeter and a standard deviation of 0.00031 millimeter, and Y is normally distributed with a mean of 0.23 millimeter and a standard deviation of 0.00017 millimeter. The value of ρ of these variables is equal to 0. Specifications call for a lamp to have a thickness of the ink corresponding to X in the range of 0.099535 and 0.100465 millimeter and Y in the range of 0.22966 and 0.23034 millimeter. What is the probability that a randomly selected lamp will conform to specifications?

Answer

0.8270